```
function bounds (marking)
  local markings = reachable (marking)
  local bound    = {
    up      = { minimum = math.huge, maximum = 0 },
    down    = { minimum = math.huge, maximum = 0 },
    fuel    = { minimum = math.huge, maximum = 0 },
    off     = { minimum = math.huge, maximum = 0 },
    on      = { minimum = math.huge, maximum = 0 },
    filled  = { minimum = math.huge, maximum = 0 },
    maximum = { minimum = math.huge, maximum = 0 },
  }
  for marking in pairs (markings) do
    bound.up.minimum = math.min (bound.up.minimum, marking.up)
    bound.up.maximum = math.max (bound.up.maximum, marking.up)
    ...
  end
  return bound
end
```